

Question	Answer	Marks
6(a)	force proportional to product of charges and inversely proportional to the square of the separation	M1
	reference to point charges	A1
6(b)(i)	(near to each sphere,) fields are in opposite directions or point (between spheres) where fields are equal and opposite or point (between spheres) where field strength is zero	M1
	so same (sign of charge)	A1
6(b)(ii)	(at $x = 5.0 \text{ cm}$,) $E = 3.0 \times 10^3 \text{ V m}^{-1}$ and $a = qE / m$	C1
	$E = (1.60 \times 10^{-19} \times 3.0 \times 10^3) / (1.67 \times 10^{-27})$	C1
	$= 2.9 \times 10^{11} \text{ m s}^{-2}$	A1
6(c)	field strength or E is potential gradient or field strength is rate of change of (electric) potential	M1
	(field strength) maximum at $x = 6 \text{ cm}$	A1